```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from google.colab import files
uploaded = files.upload()

df = pd.read_csv('House Price India (1).csv')

df.head()
```

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```
Saving House Price India.csv to House Price India (2).csv
```

	id	Date	number of bedrooms	number of bathrooms	living area	lot area	number of floors	waterfront present	number of views	condition of the house	...	Built Year	Renovation Year	Postal Code	Lattitude
0	6762810145	42491	5	2.50	3650	9050	2.0	0	4	5	...	1921	0	122003	52.8645
1	6762810635	42491	4	2.50	2920	4000	1.5	0	0	5	...	1909	0	122004	52.8878
2	6762810998	42491	5	2.75	2910	9480	1.5	0	0	3	...	1939	0	122004	52.8852
3	6762812605	42491	4	2.50	3310	42998	2.0	0	0	3	...	2001	0	122005	52.9532
4	6762812919	42491	3	2.00	2710	4500	1.5	0	0	4	...	1929	0	122006	52.9047

5 rows × 23 columns

```
df.columns
```

```
Index(['id', 'Date', 'number of bedrooms', 'number of bathrooms',
       'living area', 'lot area', 'number of floors', 'waterfront present',
       'number of views', 'condition of the house', 'grade of the house',
       'Area of the house(excluding basement)', 'Area of the basement',
       'Built Year', 'Renovation Year', 'Postal Code', 'Lattitude',
       'Longitude', 'living_area_renov', 'lot_area_renov',
       'Number of schools nearby', 'Distance from the airport', 'Price'],
      dtype='object')
```

Q1. What is the average, median, and standard deviation of house prices?

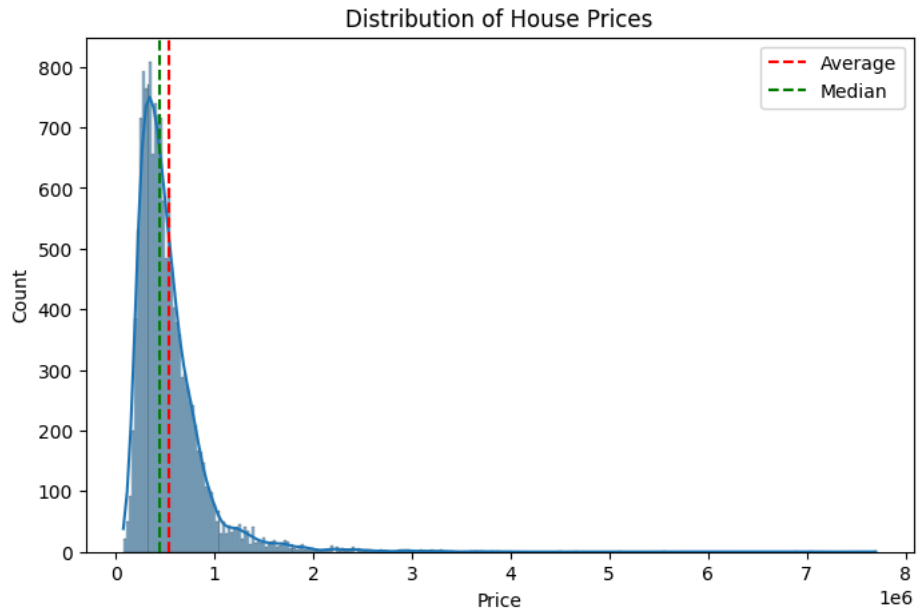
```
avg_price = df['Price'].mean()
median_price = df['Price'].median()
std_price = df['Price'].std()

print("Average Price:", avg_price)
print("Median Price:", median_price)
print("Standard Deviation:", std_price)
```

```
Average Price: 538932.2183310534
Median Price: 450000.0
Standard Deviation: 367532.3808039669
```

Visualization

```
plt.figure(figsize=(8,5))
sns.histplot(df['Price'], kde=True)
plt.axvline(avg_price, color='red', linestyle='--', label='Average')
plt.axvline(median_price, color='green', linestyle='--', label='Median')
plt.title("Distribution of House Prices")
plt.legend()
plt.show()
```



Detailed Interpretation

The average house price represents the mean market price of all houses in the dataset. The median price represents the middle value and is useful because it is less affected by extremely expensive properties. The standard deviation indicates how spread out house prices are from the average. A high standard deviation suggests that the dataset contains houses across multiple price segments, ranging from affordable to luxury properties.

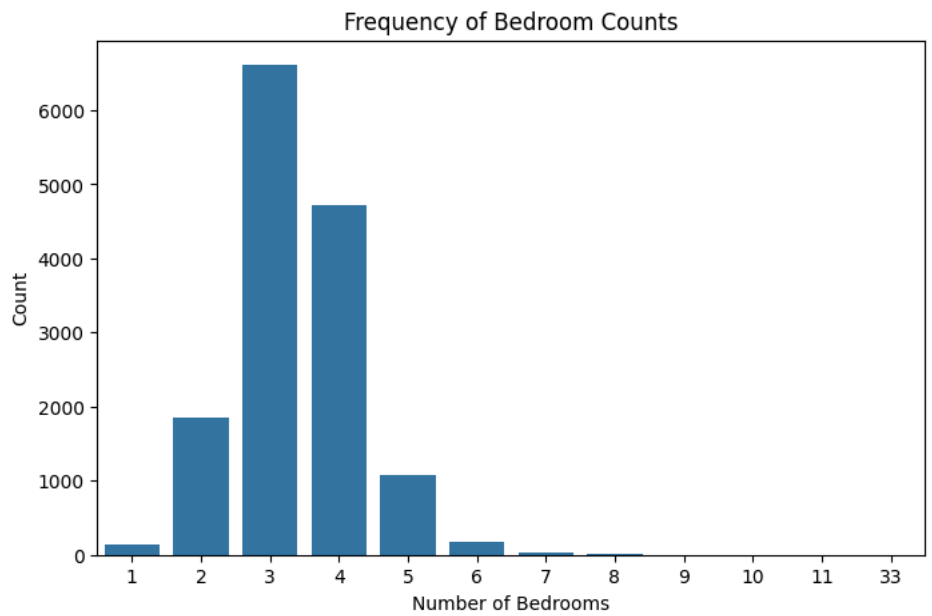
Q2. Which number of bedrooms is most common?

```
most_common_bedrooms = df['number of bedrooms'].mode()[0]
print("Most Common Bedrooms:", most_common_bedrooms)
```

```
Most Common Bedrooms: 3
```

Visualization

```
plt.figure(figsize=(8,5))
sns.countplot(x='number of bedrooms', data=df)
plt.title("Frequency of Bedroom Counts")
plt.xlabel("Number of Bedrooms")
plt.ylabel("Count")
plt.show()
```



Detailed Interpretation

The mode identifies the most frequently occurring number of bedrooms in the dataset. This helps determine the most popular house configuration among buyers. A concentration around a specific bedroom count suggests market demand is strongest for that segment.

Q3. Check if the price data is skewed.

```
skewness = df['Price'].skew()
print("Skewness:", skewness)
```

```
Skewness: 4.269297720707116
```

Visualization

```
plt.figure(figsize=(8,5))
sns.histplot(df['Price'], kde=True)
plt.title("Skewness of House Price Distribution")
plt.show()
```

Detailed Interpretation

Skewness measures the asymmetry of the price distribution. If skewness is positive, the distribution is right-skewed, meaning a small number of luxury properties significantly increase the average price. If skewness is close to zero, the prices are normally distributed. A highly skewed distribution suggests presence of outliers in premium market segments.

Q4. What is the average price for each number of bedrooms?

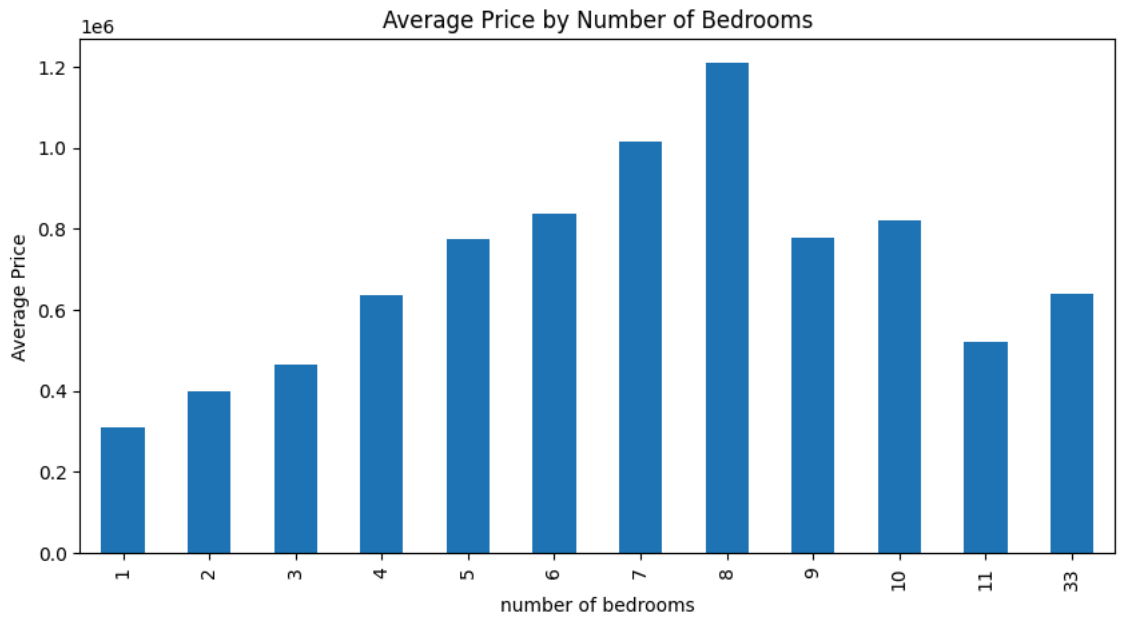
```
bedroom_price = df.groupby('number of bedrooms')['Price'].mean()
print(bedroom_price)
```

```
number of bedrooms
1     3.089638e+05
2     3.985476e+05
3     4.632776e+05
4     6.361988e+05
5     7.752550e+05
6     8.375815e+05
7     1.016544e+06
8     1.208455e+06
9     7.766663e+05
10    8.200000e+05
11    5.200000e+05
33    6.400000e+05
Name: Price, dtype: float64
```

Visualization

```
plt.figure(figsize=(10,5))
bedroom_price.plot(kind='bar')
plt.title("Average Price by Number of Bedrooms")
```

```
plt.ylabel("Average Price")
plt.show()
```



Detailed Interpretation

This analysis shows how pricing changes with bedroom count. Generally, houses with more bedrooms have higher average prices because they offer more usable space and target larger families. However, irregularities may occur if luxury small-bedroom homes exist in premium locations.

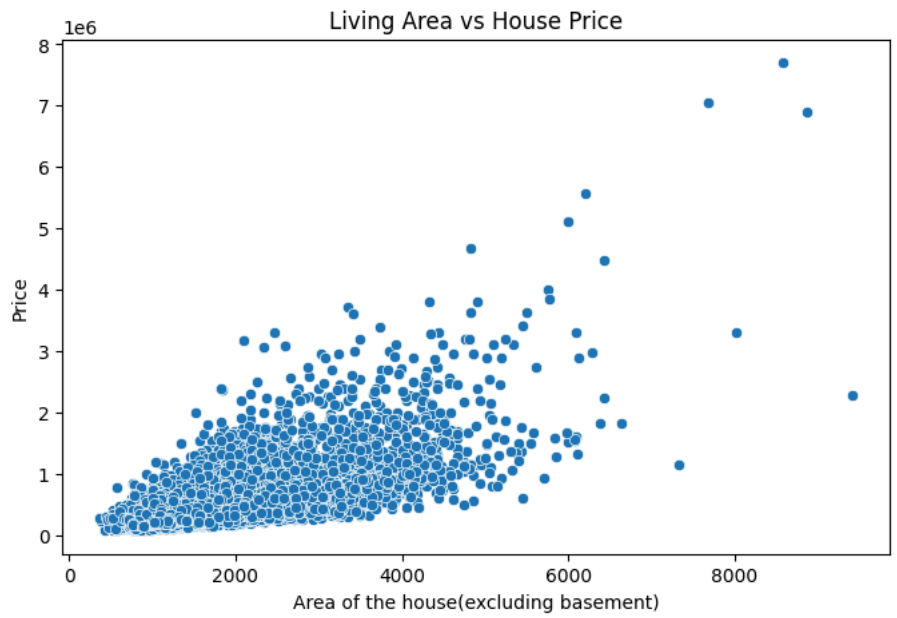
Q5. What is the relationship between living area and price?

```
correlation = df['Area of the house(excluding basement)'].corr(df['Price'])
print("Correlation:", correlation)
```

```
Correlation: 0.6152204198678108
```

Visualization

```
plt.figure(figsize=(8,5))
sns.scatterplot(
    x='Area of the house(excluding basement)',
    y='Price',
    data=df
)
plt.title("Living Area vs House Price")
plt.show()
```



Detailed Interpretation

The scatter plot and correlation value help evaluate the relationship between house size and price. A positive correlation indicates that larger houses generally command higher prices. If the correlation is strong, area is a major pricing factor in this market.

Q6. Identify anomalies where houses have high prices but low areas.

```
anomalies = df[
    (df['Price'] > df['Price'].quantile(0.90)) &
    (df['Area of the house(excluding basement)'] < df['Area of the house(excluding basement)'].quantile(0.25))
]

anomalies[['Price','Area of the house(excluding basement)']].head()
```

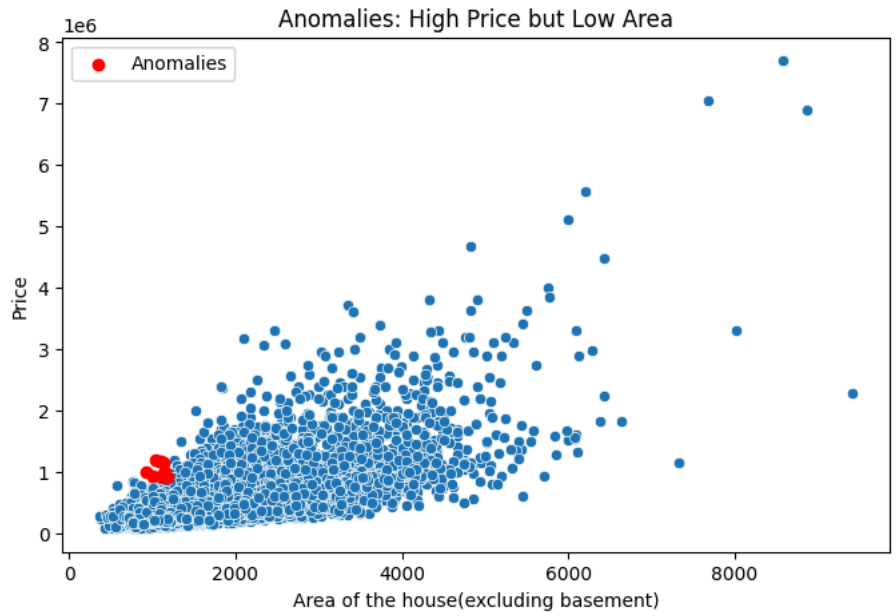
	Price	Area of the house(excluding basement)
1125	970000	1130
1210	1180000	1090
1517	1180000	1100
2154	1150000	1130
8517	990000	930

Visualization

```
plt.figure(figsize=(8,5))
sns.scatterplot(
    x='Area of the house(excluding basement)',
    y='Price',
    data=df
)

plt.scatter(
    anomalies['Area of the house(excluding basement)'],
    anomalies['Price'],
    color='red',
    label='Anomalies'
)

plt.legend()
plt.title("Anomalies: High Price but Low Area")
plt.show()
```



Detailed Interpretation

These properties appear unusual because they have relatively small living areas but exceptionally high prices. Possible reasons include:

- Premium location
- Waterfront access
- Luxury interiors

- Strong neighborhood reputation
- Commercial/resale premium

Q7. Compare average price based on number of floors and waterfront presence.

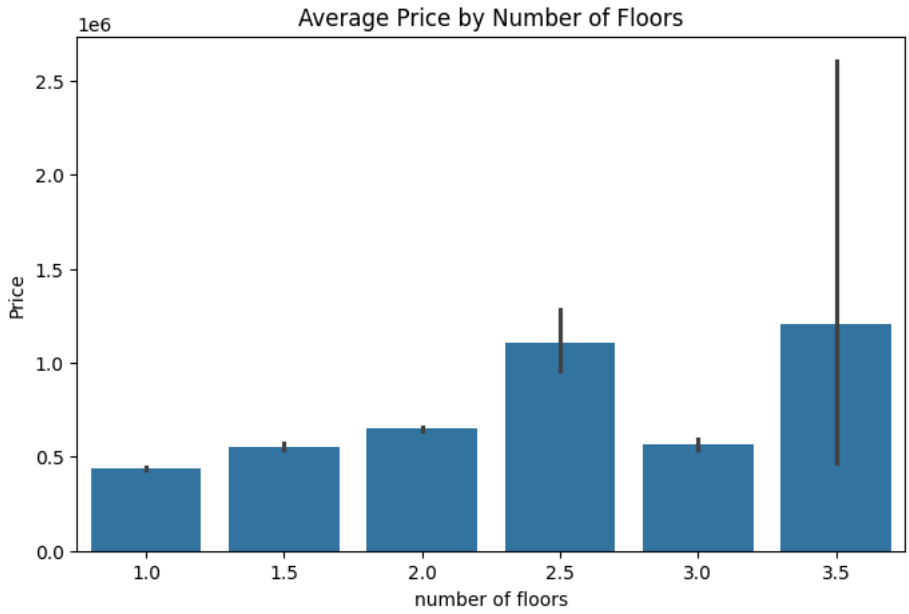
```
floor_price = df.groupby('number of floors')['Price'].mean()
waterfront_price = df.groupby('waterfront present')['Price'].mean()

print(floor_price)
print(waterfront_price)
```

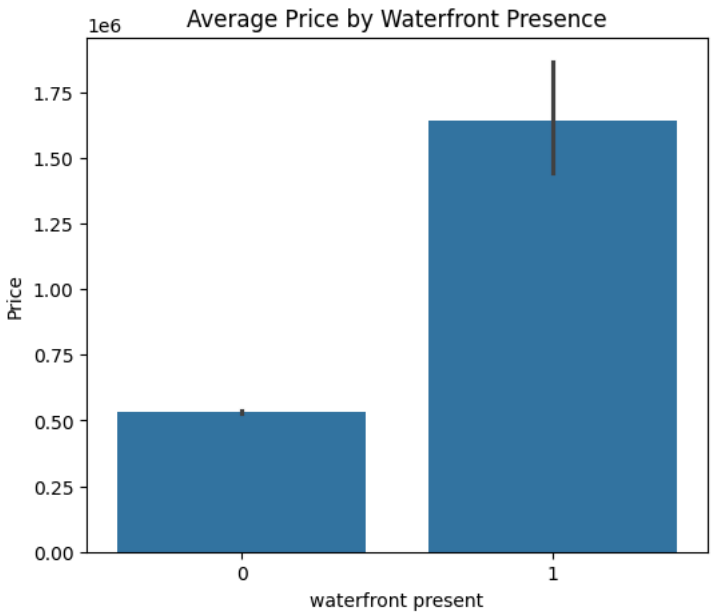
```
number of floors
1.0    4.369769e+05
1.5    5.549301e+05
2.0    6.487372e+05
2.5    1.106233e+06
3.0    5.663281e+05
3.5    1.205875e+06
Name: Price, dtype: float64
waterfront present
0    5.304174e+05
1    1.641902e+06
Name: Price, dtype: float64
```

Visualization

```
plt.figure(figsize=(8,5))
sns.barplot(x='number of floors', y='Price', data=df)
plt.title("Average Price by Number of Floors")
plt.show()
```




```
plt.figure(figsize=(6,5))
sns.barplot(x='waterfront present', y='Price', data=df)
plt.title("Average Price by Waterfront Presence")
plt.show()
```



Detailed Interpretation

Properties with more floors often have larger built-up area and may be priced higher. Waterfront homes generally command a premium due to superior location and scenic value. A large pricing difference indicates waterfront is a major luxury factor.

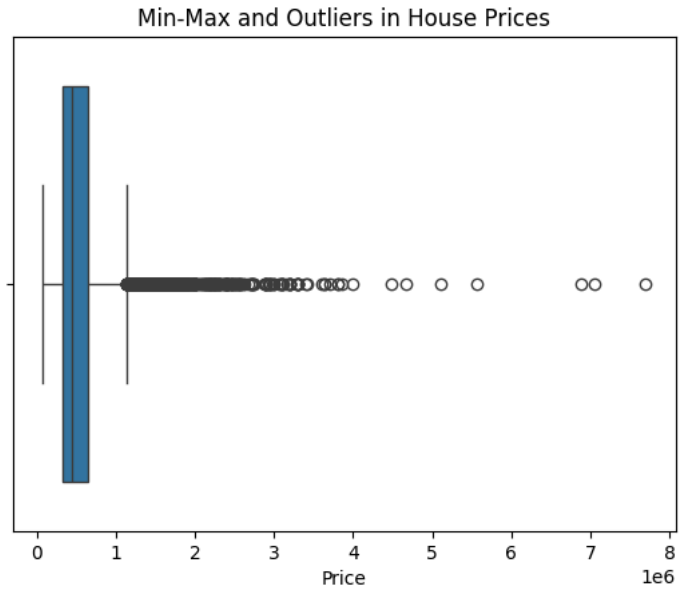
Q8. Identify minimum and maximum house price. What does this indicate?

```
print("Minimum Price:", df['Price'].min())
print("Maximum Price:", df['Price'].max())
```

```
Minimum Price: 78000
Maximum Price: 7700000
```

Visualization

```
sns.boxplot(x=df['Price'])
plt.title("Min-Max and Outliers in House Prices")
plt.show()
```



Detailed Interpretation

The wide difference between minimum and maximum house price indicates strong market segmentation. This suggests the dataset includes both affordable and premium/luxury housing categories. Outliers may represent ultra-premium properties.

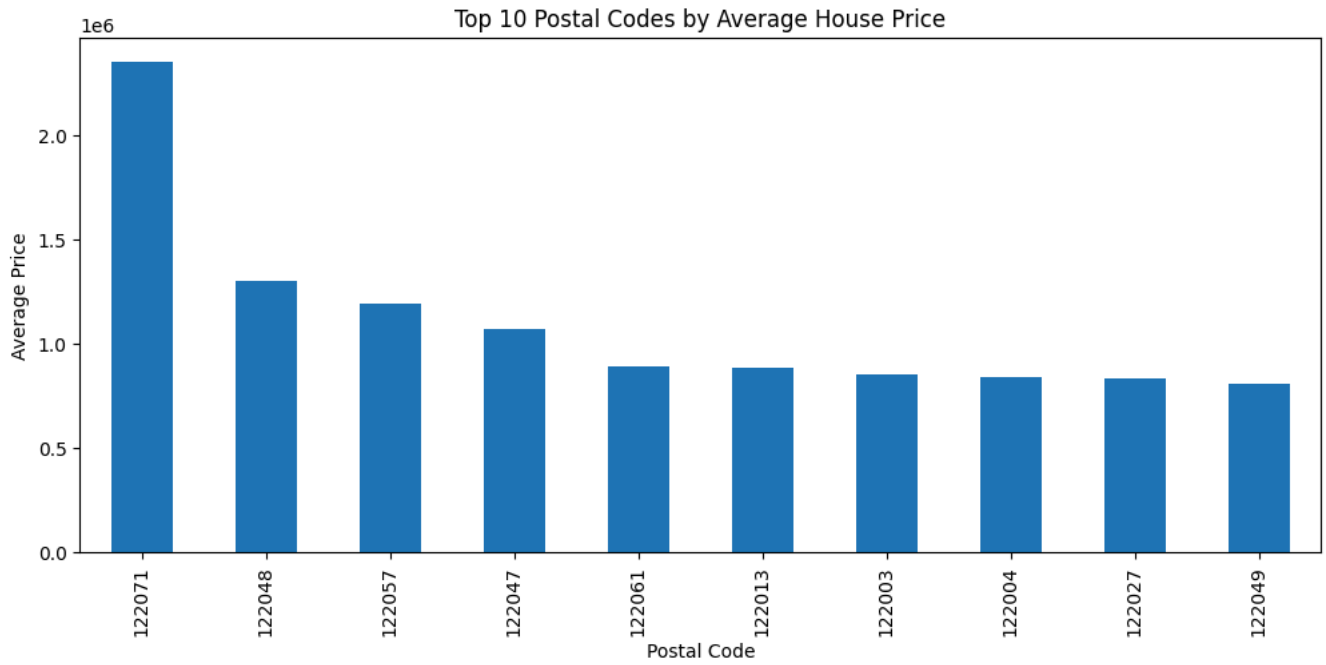
Q9. Which location (Postal Code) has the highest average price?

```
location_price = df.groupby('Postal Code')['Price'].mean().sort_values(ascending=False)
print(location_price.head())
```

```
Postal Code
122071    2.348311e+06
122048    1.296414e+06
122057    1.188517e+06
122047    1.069295e+06
122061    8.895896e+05
Name: Price, dtype: float64
```

Visualization

```
plt.figure(figsize=(12,5))
location_price.head(10).plot(kind='bar')
plt.title("Top 10 Postal Codes by Average House Price")
plt.ylabel("Average Price")
plt.show()
```



Detailed Interpretation

Postal codes with the highest average prices indicate premium residential locations. These areas may have better infrastructure, connectivity, safety, and lifestyle amenities.

Q10. Final Business Insights